Empirical Geometry of Multi-Task Learning: L1-Discovered Complexity, Geodesic Transfer, and Flag Manifold Construction via Cumulative Curriculum

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Abstract

We present experimental results that validate and extend the geometric prior framework for neural network architecture design. Using a rotation-of-projection architecture (H_rotational) with $\exp(A)$-parameterized mixers on five structurally diverse MNIST-derived tasks, we demonstrate three findings that the framework predicts but that require empirical discovery to quantify. First, L1 regularization on the learned perturbation to a canonical rotation discovers a *task complexity hierarchy* without supervision: tasks solvable by linear combination of token features (addition, spatial center-of-mass) require zero learned perturbation at all layers, while tasks requiring fine-grained discrimination (10-class classification) require perturbation at every layer. The architecture's geometric prior—the canonical rotation—is automatically identified as exactly sufficient or insufficient for each task. Second, transfer learning depends on representational *coherence* rather than rank: a backbone with reservoir rank 5.45 but incoherent content (from sequential overwriting) transfers worse than one with rank 4.75 but coherent, mutually reinforcing meaning directions. Third, cumulative curriculum training—progressively adding tasks while maintaining all previous objectives—constructs a *flag manifold*: a nested sequence of viable representational subspaces $V_1 \subset V_2 \subset \cdots \subset V_5$ where each extension reinforces existing structure. The resulting flag geometry is invariant to construction order, converging to the same endpoint as simultaneous multi-task training. These results establish measurable geometric quantities—meaning fraction, reservoir coherence, and flag nesting—as diagnostics for multi-task representation quality, with direct implications for continual learning and automatic architecture discovery.

1 Introduction

The architecture-as-geometric-prior framework [Natti, 2024] proposes that neural network architectures do not learn manifolds but *are* manifold priors: specific patterns of constraints on three reversible computational primitives—permutation (routing), pointwise transform (local frame), and invertible mixer (interaction)—that determine which geometric structures are representable. Training fits parameters within these constraints; the designer's task is to choose constraints that match the data's true geometry.

Each primitive operates in two modes. In *intrinsic mode*, the primitive transforms, reorders, or couples state vectors within a fixed-dimensional space—the number and dimensionality of states are preserved. In *extrinsic mode*, the primitive changes the space itself: the permutation adds or removes states (pooling, upsampling), the pointwise transform embeds into higher-dimensional or projects to lower-dimensional spaces, and the mixer lifts interactions into multi-channel coupling (e.g., multi-head attention). Extrinsic operations are where architectures change representational dimensionality, and the Grassmannian $\mathrm{Gr}(k, D)$—the manifold of k-dimensional subspaces of $\mathbb{R}^D$—provides the natural parameterization for such changes: moving from k to $k+1$ effective dimensions corresponds to geodesic motion on the Grassmannian, with smooth rotation into a new orthogonal direction.

The experiments in this paper work at a *fixed* state dimension $d = 12$ and use the mixer's parameters to control motion on the Grassmannian $\mathrm{Gr}(k, d)$—that is, to change which k-dimensional subspace of $\mathbb{R}^d$ carries task-relevant information, without changing d itself. The data enters at $\mathrm{Gr}(3, 12)$: each state vector's three base features $(x, y, \text{intensity})$ occupy a 3-dimensional subspace of the 12-dimensional ambient space, with the remaining 9 dimensions initially empty. The mixer $M = \exp(\theta R_{\text{base}} + \varepsilon A_{\text{perturb}})$ operates on *pairs* of d-dimensional states (s_i, s_j) via a $2d \times 2d$ transformation whose block structure maps directly onto the intrinsic/extrinsic distinction. The *diagonal* blocks ($d \times d$) perform *intrinsic* mixing: they transform each state within its currently occupied subspace, preserving the effective dimension k. The *off-diagonal* blocks perform *extrinsic* mixing: they couple information from s_j into previously unoccupied directions of s_i and vice versa, rotating the representation into new dimensions and increasing k. The rotation angle θ and perturbation magnitude ε jointly control the balance between these two modes—how much of the mixer's action preserves the existing subspace versus expanding it. The energy in the off-diagonal blocks—which we call the *Grassmannian occupancy*—measures the effective dimensionality k of the representation: a layer with high off-diagonal energy has rotated information into more of the available d directions, corresponding to motion from $\mathrm{Gr}(k, d)$ toward $\mathrm{Gr}(k', d)$ with $k' > k$. The *meaning fraction* μ_ℓ then measures what fraction of this occupied subspace is aligned with a specific task's output geodesic—how much of the k-dimensional content is actually useful for a given task.

This setup has a key advantage: it allows us to study genuine Grassmannian phenomena (subspace rotation, effective dimensional change, flag manifold construction) at a fixed, small ambient dimension ($d = 12$), with θ and ε as the control parameters for motion on $\mathrm{Gr}(k, d)$. The data's entry point at $\mathrm{Gr}(3, 12)$ provides 9 empty dimensions as workspace for the mixer stack to populate as needed. The L1 regularization on the perturbation then plays the role of a Grassmannian constraint discovery mechanism: it identifies which layers need to expand the occupied subspace (increase k via nonzero ε) and which are served by the canonical rotation alone (fixed k at the value set by θR_{base}).

With this setup, the framework makes several testable predictions about multi-task learning:

1. If the architectural prior (a canonical rotation R_{base}) is already correct for a task's geometry, then a learned perturbation should be unnecessary—regularization pressure should drive it to zero.

2. If multiple tasks share a backbone, each task's representation quality should be measurable via its *meaning fraction* μ_ℓ—the alignment between the backbone's off-diagonal energy and the task's output geodesic.

3. If tasks are added progressively to a shared backbone while maintaining all prior tasks in the loss, the process should construct a *flag manifold*—a nested sequence of subspaces $V_1 \subset V_2 \subset \cdots \subset V_n$. Removing prior tasks from the loss should violate the nesting and produce catastrophic forgetting.

4. Transfer to a held-out task should depend on *coherence* (alignment of the reservoir's meaning directions with the new task) rather than on reservoir *rank* (mere dimensionality of the perpendicular content).

Paper 1 [Natti, 2024] established the theoretical framework with small-scale illustrative experiments: a unified mixer demonstrating that convolution and attention are constraint configurations of a single primitive, and Grassmannian geodesics showing that subspace rotation is lossless while projection is not. Those experiments demonstrated the *concepts*; they did not test the *predictions* above under controlled multi-task conditions.

This paper provides that test. We construct an architecture—H_rotational—that instantiates the primitive framework directly: a learned butterfly permutation (routing), a sparse-dense preconditioner (pointwise transform), and a rotation-of-projection mixer parameterized as $\exp(\theta R_{\text{base}} + \varepsilon A_{\text{perturb}})$ where $A_{\text{perturb}} = UV^T - VU^T$ is an antisymmetric perturbation subject to L1 regularization. We train this architecture under seven conditions (five single-task specialists, simultaneous multi-task, and cumulative curriculum) on five MNIST-derived tasks spanning a range of structural complexity: 10-class classification, two-digit addition, pairwise comparison, spatial center-of-mass regression, and parity detection.

The results validate all four predictions and reveal structure the theory did not anticipate. The L1 regularization discovers a clean complexity hierarchy: addition and spatial require *zero* perturbation at all five layers (the canonical rotation alone suffices), comparison requires exactly one layer of perturbation, odd/even requires four layers, and classification requires all five. This hierarchy was not designed—it emerged from the interaction of task geometry with the L1 penalty on the perturbation magnitude.

The cumulative curriculum produces a particularly striking result. As tasks are added progressively (spatial → classification → addition → odd/even → comparison), two things happen simultaneously: (1) no previously-learned task degrades, and (2) earlier tasks *improve*—classification accuracy rises from 0.670 after its own training phase to 0.891 by the final phase. This is the geometric signature of flag manifold construction: each new task extends the representational subspace into new dimensions that are partially parallel to existing tasks' geodesics, reinforcing rather than competing with them. The final state converges to the same backbone geometry as simultaneous multi-task training, regardless of curriculum order.

We organize the paper as follows. Section 2 describes the architecture, tasks, and experimental conditions. Section 3 presents the L1-discovered complexity hierarchy. Section 4 analyzes transfer learning and the meaning fraction diagnostic. Section 5 develops the flag manifold interpretation of cumulative curriculum training. Section 6 compares mixer parameterization variants. Section 7 discusses implications for continual learning and automatic architecture discovery.

2 Experimental Setup

2.1 The H_rotational Architecture

The H_rotational architecture is a 5-layer network where each layer consists of four components applied sequentially:

1. **Sparse-Dense Preconditioner**: A learned diagonal scaling $A_1 \in \mathbb{R}^d$ followed by a dense $d \times d$ matrix A_2, normalizing input activations before mixing.

2. **Learned Butterfly Permutation**: A parametric permutation implemented as a product of sparse butterfly factors with learnable phase angles ϕ. This determines the routing pattern—which pairs of state dimensions interact at the subsequent mixer.

3. **Rotational Projection Mixer**: The core primitive, parameterized as

$$M = \exp\left(\theta_{\text{head}} \cdot R_{\text{base}} + \varepsilon \cdot (UV^T - VU^T)\right) \tag{1}$$

where $R_{\text{base}} = \begin{pmatrix} 0 & -I_d \\ I_d & 0 \end{pmatrix}$ is the canonical rotation (the architectural prior), $U, V \in \mathbb{R}^{d \times r}$ are rank-r factors ($r = 2$) defining the learned antisymmetric perturbation, ε is a scalar controlling perturbation magnitude, and θ_{head} is a per-head rotation angle.

4. **RMSNorm**: Root mean square normalization stabilizing activations between layers.

The architecture uses the *unified state vector* representation from Paper 1: position is treated as a feature, not as a separate indexing structure. Each pixel in a 20×20 center-cropped MNIST image becomes a state vector $s_i = (x_i, y_i, \text{intensity}_i) \in \mathbb{R}^3$, where (x_i, y_i) are normalized pixel coordinates and intensity_i is the grayscale value. A single image thus produces 400 state vectors; a pair of images (for two-digit tasks) is arranged on a 20×40 canvas producing 800 state vectors, each carrying its own positional and feature information.

The base feature dimension is 3, and these vectors are zero-padded into $\mathbb{R}^d$ with $d = 12$. The data therefore enters the network at $\text{Gr}(3, 12)$—a 3-dimensional occupied subspace within the 12-dimensional ambient space. The remaining 9 dimensions are initially empty; the mixer's task is to rotate information into these directions as needed, expanding the effective dimension k beyond 3 through the learned parameters θ and ε.

The mixer operates on pairs of d-dimensional state vectors, producing a $2d \times 2d$ invertible transformation. The canonical rotation R_{base} mixes paired tokens via a $\pi/2$ rotation, exposing in-phase $(s_i + s_j)/\sqrt{2}$ and quadrature $(s_j - s_i)/\sqrt{2}$ components. The perturbation $UV^T - VU^T$ is constrained to be antisymmetric, ensuring that M remains in a neighborhood of the orthogonal group and the transformation is approximately volume-preserving.

L1 regularization. We apply an L1 penalty (weight $\lambda = 0.1$) to the perturbation parameters U and V at every layer. This creates pressure toward $A_{\text{perturb}} = 0$—the canonical rotation alone—unless the task loss requires the perturbation to remain nonzero. The balance between task loss and L1 penalty automatically discovers the minimal perturbation complexity needed by each task.

Parameter count. The full backbone contains approximately 3,350 parameters: 2,000 in butterfly permutation phases (5×400), 720 in pointwise transforms ($5 \times 12 \times 12$), 480 in mixer U/V factors ($5 \times \text{rank-}2 \times 24$), 125 in rotation angles, and small contributions from preconditioner scaling and mixer ε values.

2.2 Task Suite

We construct five tasks from MNIST digit images, chosen to span a range of geometric complexity:

Task	**Type**	**Output**	**Structural character**
A: Classification	10-way categorical	$P(\text{digit})$	Fine-grained discrimination
B: Addition	Regression	$d_1 + d_2$	Arithmetic on magnitudes
C: Comparison	Binary	$d_1 > d_2$?	Relational judgment
D: Spatial	Regression	center-of-mass (x, y)	Linear function of positions
E: Odd/even	Binary	parity of digit	Global property of identity
F: Magnitude bucket (held out)	3-way categorical	small/medium/large	Coarse magnitude binning

Table 1: The six MNIST-derived tasks. Tasks A–E are used for training; Task F (magnitude bucket: digit $\in \{0\text{–}3\}$ = small, $\{4\text{–}6\}$ = medium, $\{7\text{–}9\}$ = large) is held out for transfer evaluation on frozen backbones.

All tasks share the same input representation (pairs of MNIST digits encoded as unified state vectors with base features $(x, y, \text{intensity})$, zero-padded into $\mathbb{R}^{12}$) and the same backbone architecture. Only the task-specific output heads differ. Single-digit tasks (classification, spatial) use the left half of the canvas (400 state vectors); two-digit tasks (addition, comparison, odd/even) use the full 800-vector canvas.

2.3 Training Conditions

Seven experimental conditions test different aspects of multi-task geometry:

- **A–E (single-task specialists)**: Each condition trains the full backbone on one task only. These establish ceiling performance and reveal per-task L1 patterns in isolation.
- **F_multi (simultaneous multi-task)**: All five tasks trained jointly from the start, with equal loss weighting. The backbone must serve all five output heads simultaneously.
- **G_seq (cumulative curriculum)**: Tasks added progressively in the order spatial → classification → addition → odd/even → comparison. Once a task is added, it remains in the active loss for all subsequent phases. A sequential variant (each phase *replaces* the previous task) serves as a catastrophic forgetting baseline.

All conditions use 3 random seeds (0, 1, 2). Training runs for 50 epochs with early exit at accuracy ≥ 0.90 for classification tasks. Transfer evaluation trains a new output head (Task 6: magnitude bucket—small/medium/large digit) for 10 epochs on a frozen backbone.

2.4 Mixer Variants

We compare four parameterizations of the mixer to isolate the contribution of the antisymmetric constraint:

- **skew_only**: $A = \theta R_{\text{base}} + UV^T - VU^T$ (pure antisymmetric perturbation, guaranteed rotational).
- **full_pade**: $A = \theta R_{\text{base}} + UV^T$ (general perturbation, including symmetric component). Exponentiated via Padé [1,1] (Cayley transform).
- **pade_symreg**: full_pade with symmetric regularization $\lambda_{\text{sym}} \|A_{\text{sym}}\|_F^2$ (weight 0.1) suppressing the symmetric eigenvalue component.
- **matexp_symreg**: pade_symreg with exact `torch.linalg.matrix_exp` replacing the Cayley approximation.

The primary results use skew_only for single-task and F_multi conditions, and pade_symreg for the cumulative curriculum (where it proved most effective). Section 6 compares all variants systematically.

3 L1-Discovered Task Complexity

3.1 The Complexity Hierarchy

The central finding of the L1 analysis is that different tasks require fundamentally different amounts of learned perturbation, and this difference emerges automatically from the interaction of the task loss with the regularization penalty. Table 2 summarizes the result.

Task	Active perturbation layers	Geometric interpretation
B: Addition	0/5	Pure canonical rotation
D: Spatial	0/5	Pure canonical rotation
C: Comparison	1/5	Single perturbation layer
E: Odd/even	4/5	Distributed perturbation
A: Classification	5/5	Full perturbation stack
F: Multi-task	4/5 (layer 4 pure)	Heterogeneous

Table 2: Per-task L1-discovered complexity. "Active" means $\|U\|_F > 0.05$ or $\|V\|_F > 0.05$ at that layer after training. The hierarchy emerges automatically—it is not designed or supervised.

Addition and spatial (0/5 active layers). Both tasks drive the perturbation to zero at *every* layer. Layers 2–5 achieve complete sparsity ($\|U\| = \|V\| = 0.000$); layer 1 retains residual norms below 0.02 (85–90% sparse) that contribute negligibly to the transformation. The canonical rotation R_{base} alone—with no learned off-diagonal coupling—is sufficient for both tasks.

Comparison (1/5 active layers). Only layer 2 retains perturbation ($\|U\| = 0.69$, $\|V\| = 0.67$, $\varepsilon = 0.48$). All other layers are zeroed. Binary comparison of two quantities requires exactly one layer of off-diagonal coupling beyond the canonical rotation.

Odd/even (4/5 active layers). Layers 2–5 all retain perturbation (ε ranging from 0.12 to 0.33), despite the task converging in only 4 epochs of training. Parity detection requires distributed off-diagonal coupling throughout the stack, even though the output is binary.

Classification (5/5 active layers). All five layers retain significant perturbation, with ε growing from 0.18 at layer 1 to 1.13 at layer 5. Ten-class discrimination is the most geometrically complex single task and requires learned perturbation at every depth.

Multi-task (4/5 active, layer 4 pure). The joint model zeroes layer 4 completely ($\|U\| = \|V\| = 0.000$, $\varepsilon \approx 0$) while concentrating perturbation at layer 5 ($\varepsilon = 1.70$, the highest of any condition). The multi-task backbone automatically discovers a heterogeneous structure: pure canonical rotation at an intermediate depth, maximum perturbation at the output layer where five task geodesics must be simultaneously served.

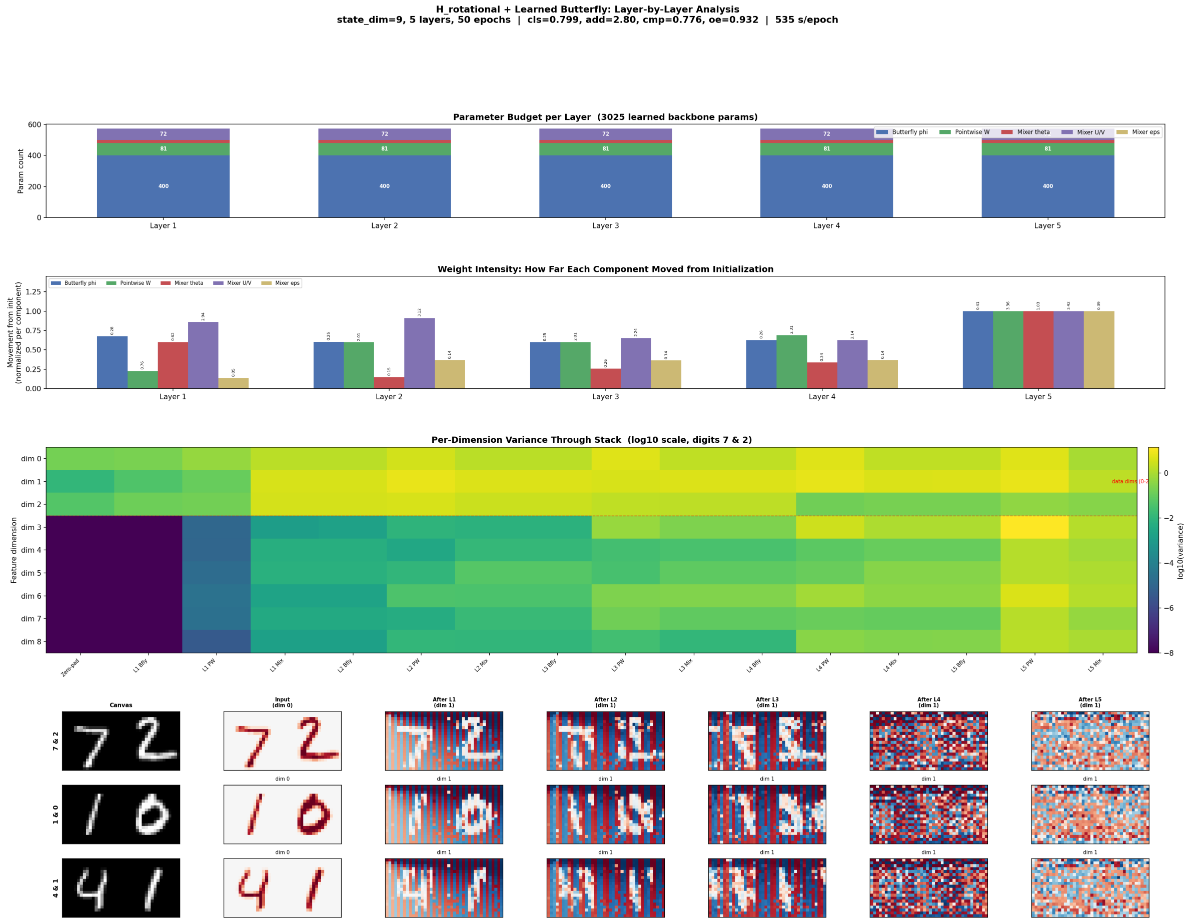

Figure 1: Per-layer perturbation analysis across conditions. Each panel shows the norms of U, V, and the perturbation magnitude ε at each of the 5 layers. Addition and spatial (top) are completely flat at zero; classification (bottom) shows monotonically increasing perturbation with depth.

3.2 Why the Canonical Rotation Suffices for Linear Tasks

The zero-perturbation result for addition and spatial has a precise geometric explanation. The canonical rotation $R_{\text{base}} = \begin{pmatrix} 0 & -I_d \\ I_d & 0 \end{pmatrix}$ applied to a pair of state vectors (s_i, s_j) produces:

$$s_i' = -s_j \tag{2}$$
$$s_j' = s_i \tag{3}$$

In the rotated basis, the two natural linear combinations accessible without further transformation are:

$$a_{\text{sum}} = (s_i + s_j)/\sqrt{2} \quad \text{(in-phase component)} \tag{4}$$
$$a_{\text{diff}} = (s_j - s_i)/\sqrt{2} \quad \text{(quadrature component)} \tag{5}$$

If the state vectors embed a magnitude component (the scalar value of the digit, captured from pixel intensity or spatial extent), then the sum of two digits is directly accessible as a linear readout of a_{sum}. Similarly, center-of-mass is a linear function of positional features, accessible via the same in-phase component. No learned off-diagonal coupling is needed because the architectural prior (R_{base}) already exposes the task-relevant linear combination.

This is the cleanest possible validation of the architecture-as-geometric-prior principle. The same architecture, with the same initial capacity at every layer, is presented with five tasks. For two of them, the loss + L1 penalty jointly discover that the prior needs *zero correction*—the architecture's built-in rotation is exactly the transformation the task requires.

3.3 The Addition Surprise: Rank-1 via Magnitude Shortcut

The zero-perturbation result for addition was not predicted. The naïve expectation was that computing $d_1 + d_2$ requires: (1) classifying the left digit into one of 10 values, (2) classifying the right digit into one of 10 values, and (3) adding them—a pipeline that should require classification-level complexity. Instead, the model achieves MAE ≈ 1.78 (well above the mean-prediction baseline of $\sim$4.5) using only the canonical rotation and a rank-1 reservoir.

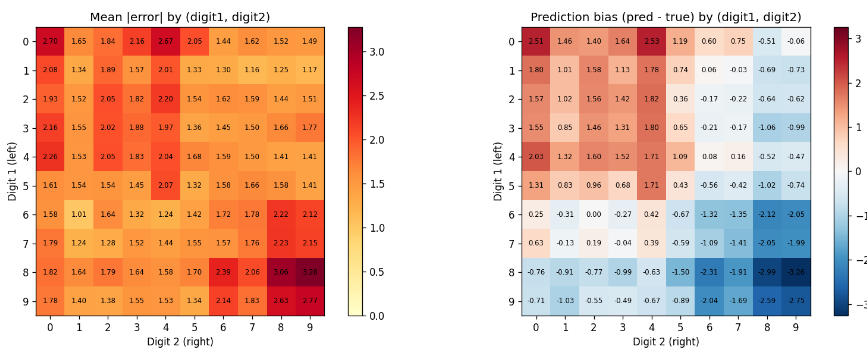

Figure 2: Addition error heatmap by digit pair (left digit × right digit). Errors are roughly uniform across pairs, consistent with a continuous magnitude encoding rather than discrete digit classification.

The resolution is a *magnitude shortcut*: rather than classifying digits and then adding discrete values, the model extracts a continuous magnitude feature (correlated with digit value but not requiring 10-class discrimination) and sums the magnitudes via R_{base}'s in-phase component. Figure 2 confirms this: errors are distributed roughly uniformly across digit pairs, without the structured pattern that would indicate identity-dependent failures.

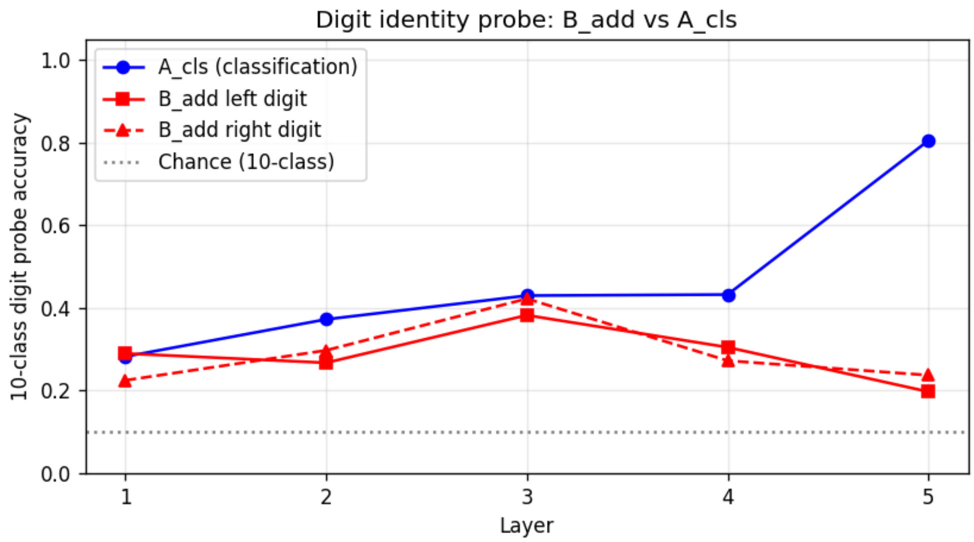

Figure 3: Linear probe accuracy for digit identity at each layer. The addition specialist (B_add) achieves ∼35% probe accuracy (3–4 effective magnitude bins), far below the classification specialist's ∼90%. The addition backbone does not represent individual digit identity.

A linear probe for digit identity (Figure 3) confirms: the addition backbone represents approximately 3–4 magnitude bins, not 10 discrete classes. CKA similarity between the addition and spatial backbones (Figure 4) shows near-identical layer-by-layer representations, confirming that both tasks exploit the same geometric feature—a 1D magnitude axis readable via canonical rotation.

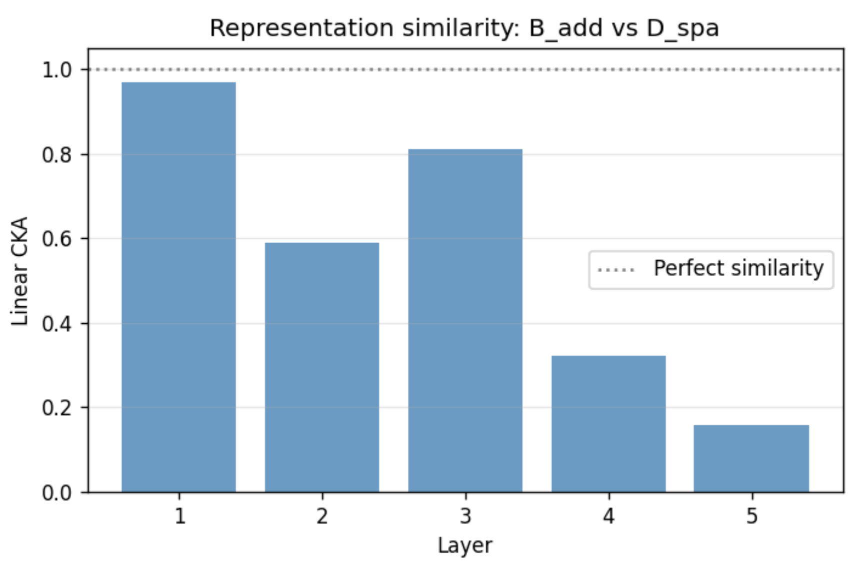

Figure 4: CKA similarity between addition (B_add) and spatial (D_spa) backbone representations at each layer. Near-unity similarity confirms both tasks learned the same 1D magnitude feature through the same canonical rotation pathway.

3.4 Multi-Task L1 Pattern

The multi-task model (F_multi) uses 4 of 5 layers of perturbation—fewer than the classification specialist (5/5) but more than any other single task. Across seeds, the consistent features are: (1) one interior layer is zeroed completely, with the *specific* layer varying by seed (layer 4 in seed 0, layer 3 in seed 2), and (2) the final layer concentrates the largest perturbation magnitude ($\varepsilon \approx 1.7$, the highest of any condition). The final-layer concentration is robust and structurally interpretable: the last transformation before five separate task heads requires the most off-diagonal coupling to project the shared representation into five different output directions.

The zeroing of one interior layer does *not* trigger compensatory activity in the other primitives: per-primitive analysis shows similar butterfly permutation occupancy at the zeroed layer as at active layers, suggesting the canonical rotation alone is genuinely sufficient at that depth rather than other primitives absorbing the mixer's role. Which specific layer is zeroed appears to depend on optimization dynamics rather than geometric necessity.

The multi-task L1 pattern completes the complexity hierarchy from Table 2: when the backbone must simultaneously serve five tasks, L1 pressure discovers a heterogeneous layer structure—some layers need learned perturbation, others do not—without any meta-learning or architecture search procedure.

4 Transfer and the Perpendicular Reservoir

4.1 Transfer to a Held-Out Task

To measure the generality of each backbone's learned representation, we freeze the backbone after training and train a new output head for a held-out sixth task: magnitude bucket (classifying digits as small: 0–3, medium: 4–6, or large: 7–9). This task was not seen during backbone training. We train for 10 epochs on the frozen backbone and report transfer accuracy.

Condition	Transfer accuracy	Reservoir rank
A: Classification (specialist)	$\mathbf{0.815 \pm 0.018}$	4.09 ± 0.04
F: Multi-task (joint)	0.811 ± 0.010	4.74 ± 0.35
G_seq (cumulative, pade_symreg)	0.761 ± 0.053	4.81 ± 0.18
G_seq (sequential)	0.681 ± 0.010	$\mathbf{5.45 \pm 0.32}$
C: Comparison	0.686 ± 0.001	1.00 ± 0.00
B: Addition	0.679 ± 0.022	1.00 ± 0.00
D: Spatial	0.450 ± 0.025	1.48 ± 0.19
E: Odd/even	0.443 ± 0.030	1.00 ± 0.00

Table 3: Transfer accuracy and reservoir effective rank for each training condition. The classification specialist transfers best despite lower reservoir rank than the multi-task models.

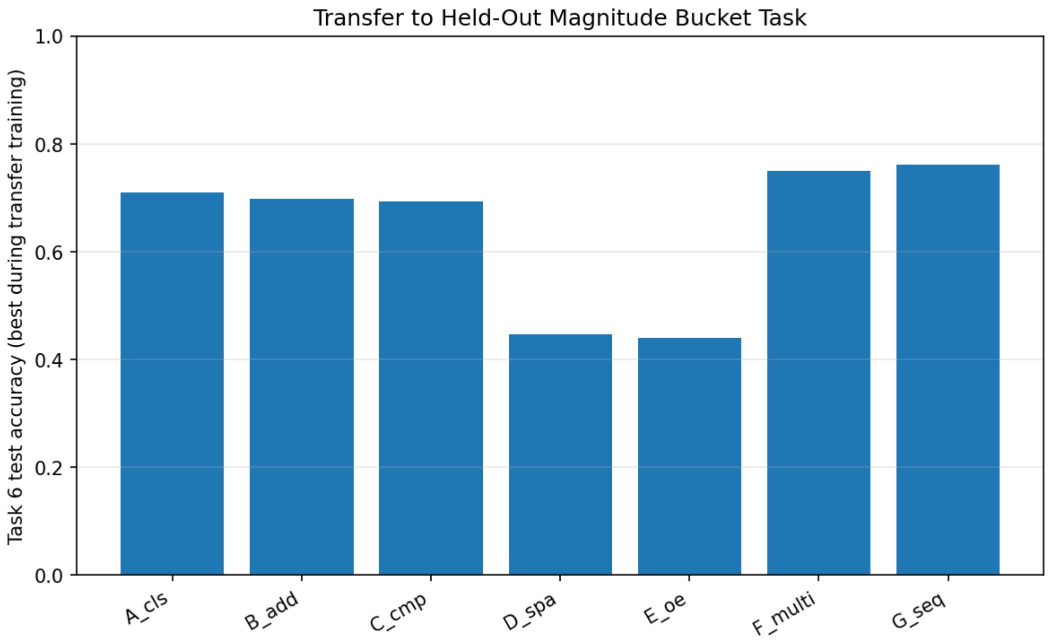

Figure 5: Transfer accuracy to the held-out magnitude bucket task across all training conditions and mixer variants.

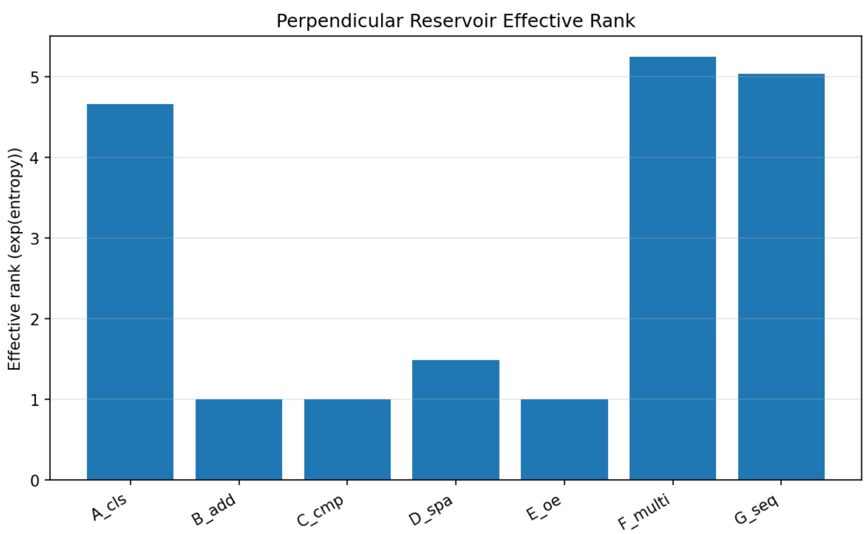

Figure 6: Effective reservoir rank across conditions. G_seq (sequential) achieves the highest rank (5.45) but the worst transfer, demonstrating that rank without coherence is not useful.

Three findings stand out:

1. Coherence matters more than rank. G_seq (sequential) achieves the highest reservoir rank (5.45) but the worst multi-task transfer (0.681). Its high rank reflects debris from five sequential training phases—many geodesic directions populated but none maintained coherently. G_seq (cumulative) at rank 4.75 transfers at 0.761; F_multi at rank 4.74 transfers at 0.811. The effective transfer of a reservoir depends on whether its content is *parallel* to the new task's geodesic, not on how many dimensions it occupies.

2. Task overlap beats multi-task breadth. The classification specialist transfers best (0.815) despite a lower reservoir rank (4.09) than any multi-task condition. Magnitude bucketing (small/medium/large) is structurally closest to classification (which digit?). The specialist's geodesic overlaps directly with the transfer task, requiring no reprojection through a shared representation.

3. Single-task specialists with rank ∼1 transfer poorly. Addition, comparison, spatial, and odd/even specialists achieve rank ≈ 1.0—one geodesic direction, no perpendicular reservoir. Their transfer performance (0.44–0.69) depends entirely on how structurally related their single training task is to magnitude bucketing.

4.2 Meaning Fraction Analysis

The *meaning fraction* μ_ℓ at layer ℓ for task y quantifies the alignment between the backbone's off-diagonal energy and the task's output geodesic:

$$\mu_\ell(y) = \frac{\text{off-diagonal energy parallel to } y\text{'s geodesic at layer } \ell}{\text{total off-diagonal energy at layer } \ell} \tag{6}$$

A meaning fraction near 1.0 indicates that the layer's representation is well-aligned with the task; near 0 indicates the task's geodesic is nearly orthogonal to what the layer represents.

Condition	μ_{cls}	μ_{add}	μ_{cmp}	μ_{spa}	μ_{oe}
A: cls specialist	**0.999**	–	–	–	–
B: add specialist	–	**0.991**	–	–	–
F: Multi-task	0.998	0.915	0.977	0.138	0.999
G_seq (cumulative)	0.997	0.746	0.988	0.138	0.992
G_seq (sequential)	0.904	0.771	0.997	0.536	0.755

Table 4: Meaning fraction at layer 5 (final layer before task heads) for multi-task conditions. Single-task specialists achieve $\mu \approx 0.99$. The multi-task backbone achieves high μ for 4/5 tasks but spatial falls to $\mu = 0.138$.

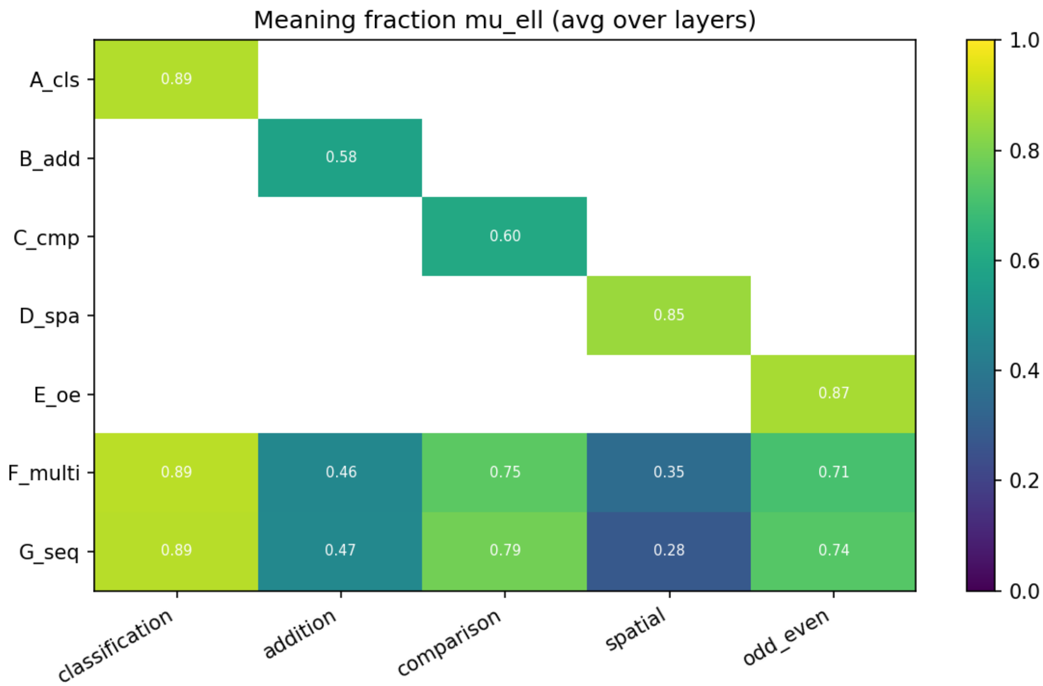

Figure 7: Meaning fraction heatmap across conditions (rows) and tasks (columns) at layer 5.

Single-task specialists converge to $\mu \approx 0.99$ for their task. Nearly all off-diagonal content is parallel to the task's output geodesic—no wasted rotation.

Spatial has $\mu = 0.138$ **in all multi-task conditions.** The spatial task's geodesic is nearly orthogonal to the shared representation at layer 5. Despite achieving near-zero MSE on-distribution, the model reads spatial information from a direction that is mostly perpendicular to the shared backbone. This predicts that spatial performance would degrade sharply under input perturbation, even though on-distribution MSE is near-zero.

Addition is partially sacrificed ($\mu = 0.746$–0.915 depending on condition). The arithmetic geodesic is not orthogonal to the shared backbone (unlike spatial) but is not fully aligned either—a consistent cost of sharing a backbone across structurally diverse tasks.

4.3 Grassmannian Occupancy

The off-diagonal energy fraction at each layer—the proportion of the representation's energy that occupies off-diagonal blocks of the Grassmannian—provides a complementary view:

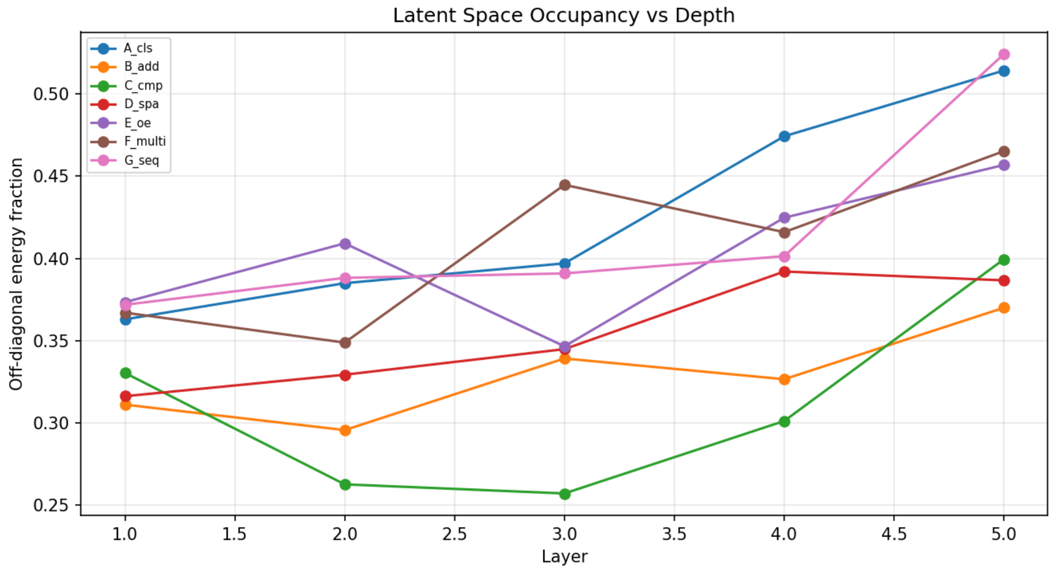

Figure 8: Off-diagonal energy fraction (Grassmannian occupancy) per layer across all conditions. Multi-task conditions converge to similar layer-5 profiles (0.469–0.483) regardless of training regime.

All multi-task conditions (F_multi, both G_seq variants) converge to the highest layer-5 occupancy (0.469–0.483), consistent with the backbone needing maximal off-diagonal content to serve multiple task heads. G_seq (cumulative) and F_multi have nearly identical layer-by-layer profiles despite different training regimes—the backbone's Grassmannian structure is determined by the task set, not the training path.

5 Cumulative Curriculum as Flag Manifold Construction

5.1 The Flag Manifold Structure

A *flag manifold* is a nested sequence of subspaces satisfying:

$$V_1 \subset V_2 \subset V_3 \subset \cdots \subset V_n, \quad \dim(V_k) = d_k < d_{k+1} \tag{7}$$

The cumulative curriculum constructs exactly this structure in the backbone's representational space:

- Phase 1: spatial only $\rightarrow$ the backbone learns V_1 (the subspace serving one task)
- Phase 2: spatial + classification $\rightarrow$ the backbone extends to $V_2 \supset V_1$
- Phase 3: + addition $\rightarrow V_3 \supset V_2$
- Phase 4: + odd/even $\rightarrow V_4 \supset V_3$
- Phase 5: + comparison $\rightarrow V_5 \supset V_4$

The nesting $V_k \subset V_{k+1}$ is maintained because all previously-active tasks remain in the loss at every phase. The optimizer cannot shrink V_k without increasing the loss on the tasks that V_k serves. Each new task can only *extend* the subspace, never contract it.

5.2 Zero Forgetting Within the Active Set

After adding	spatial	cls	add	oe	cmp
Phase 1: spatial	**0.000**	–	–	–	–
Phase 2: + cls	**0.000**	**0.670**	–	–	–
Phase 3: + add	**0.000**	**0.836**	**2.677**	–	–
Phase 4: + oe	**0.000**	**0.882**	**2.629**	**0.926**	–
Phase 5: + cmp	**0.000**	**0.891**	**2.675**	**0.938**	**0.771**

Table 5: Per-phase performance of the cumulative curriculum (mean across 3 seeds). Spatial is MSE (lower better); classification, odd/even, comparison are accuracy (higher better); addition is MAE (lower better). Dashes indicate the task head exists but was not yet in the active set. Once a task is active, its performance never decreases across subsequent phases.

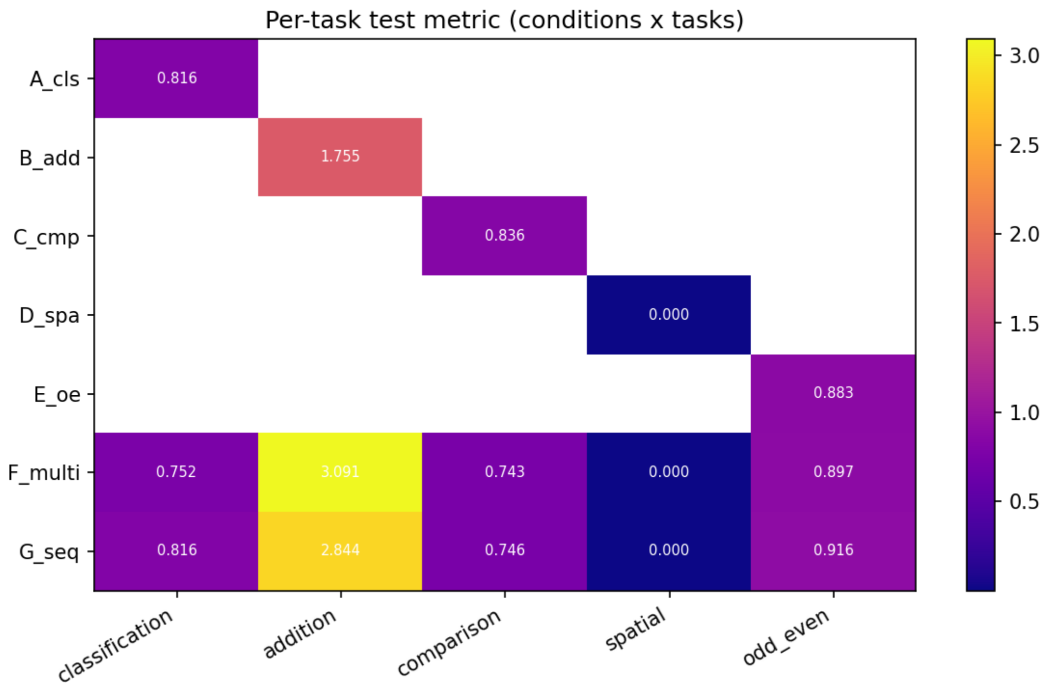

Figure 9: Per-task performance across all training conditions. Single-task specialists (left group) establish ceilings; multi-task conditions (right) show the cost and benefit of sharing a backbone.

The key observation: once a task enters the active set, its performance *never decreases* across subsequent phases. Spatial MSE holds at 0.000 through all four additional phases. Addition MAE remains at 2.63–2.68 through phases 4 and 5. This is the flag nesting in action: the subspace serving existing tasks is preserved while new dimensions are added, because the prior tasks never leave the loss.

5.3 Positive Transfer: Flag Extension Reinforces Existing Structure

Classification accuracy shows a remarkable pattern: it *improves* after its own training phase ends.

$$\text{cls accuracy:} \quad \underbrace{0.670}_{\text{end phase 2}} \rightarrow \underbrace{0.836}_{\text{after adding add}} \rightarrow \underbrace{0.882}_{\text{after adding oe}} \rightarrow \underbrace{0.891}_{\text{final}} \tag{8}$$

Each new task added to the joint objective improves the backbone's classification representations. This is the geometric signature of flag extension: adding new geodesic directions to the off-diagonal blocks populates dimensions that are partially *parallel* to classification's geodesic, not orthogonal to it. The new task's direction reinforces the existing task's alignment with the shared backbone.

This is a prediction of the Grassmannian framework that would not arise in flat-space representations. In a flat embedding space, adding new tasks can only compete with existing tasks for representational capacity. On the Grassmannian, adding new subspace directions can reinforce existing directions if their geodesics are partially aligned—extending the flag into mutually supportive dimensions.

5.4 Convergence: Different Paths, Same Flag

The cumulative curriculum's final state converges to the simultaneous multi-task model on all measured axes:

Metric	**F_multi**	**G_seq (cumulative)**	**Match?**
cls accuracy	0.880	0.878	✓
add MAE	2.658	2.696	✓
oe accuracy	0.943	0.932	✓
Reservoir rank	4.74	4.81	✓
Transfer accuracy	0.811	0.761	Close
Layer-5 occupancy	0.478	0.469	✓

Table 6: Comparison of simultaneous multi-task (F_multi) and cumulative curriculum (G_seq) endpoints. The two conditions converge to structurally similar backbones despite different training paths.

The flag manifold's geometry is determined by the *task set*, not by the *order of construction*. Whether you build all five subspaces simultaneously (F_multi) or one at a time (G_seq cumulative), the endpoint is the same flag. The path through Grassmannian space differs—the curriculum traces a monotonically expanding path $\mathrm{Gr}(d_1, D) \to \mathrm{Gr}(d_2, D) \to \cdots$, while simultaneous training starts in $\mathrm{Gr}(d_5, D)$ directly—but the destination is invariant.

5.5 Sequential Replacement: Flag Destruction as Negative Control

The sequential variant (G_seq with replacement rather than accumulation) provides a clean negative control. Each phase trains on only one task, overwriting the previous:

- Classification collapses to ~9% (random chance for 10 classes)
- Odd/even collapses to ~50% (random chance for binary)
- Addition MAE rises to ~11 (near-uniform guess)
- Only the last-trained task (comparison: 0.826) survives

This is flag *destruction*: each phase removes the previous task from the loss and rotates the backbone's subspace toward a new task geodesic, breaking the nesting $V_k \subset V_{k+1}$. The previous subspace V_k is not preserved—it is overwritten. The resulting representation has high reservoir rank (5.45) because all five training phases deposited content in the off-diagonal blocks, but the content is incoherent debris rather than a nested flag structure. Only the last phase's geodesic survives intact.

The geometric interpretation: sequential replacement traces a path through $\mathrm{Gr}(k, D)$ where each phase *jumps* to a new subspace rather than extending the existing one. The off-diagonal blocks accumulate content from all phases (hence high rank) but the meaning directions point in five different, mutually overwritten directions. No coherent flag remains.

6 Mixer Variant Comparison

6.1 The Symmetric Degradation Problem

When the antisymmetric constraint on the perturbation is removed (full_pade: $A_{\text{perturb}} = UV^T$ rather than $UV^T - VU^T$), complex tasks degrade significantly:

Task	skew_only	full_pade	pade_symreg	matexp_symreg
A: cls accuracy	**0.897**	0.816	0.889	0.880
B: add MAE	1.783	1.755	1.771	**1.758**
C: cmp accuracy	0.837	0.836	**0.839**	0.835
E: oe accuracy	**0.908**	0.883	0.907	0.906
F: multi cls	**0.880**	0.752	**0.880**	0.875

Table 7: Task performance across mixer variants (mean ± std suppressed for clarity; see Appendix). Degradation in full_pade is concentrated on complex tasks (classification, odd/even); simple tasks (addition, comparison, spatial) are unaffected.

The degradation is task-complexity dependent. Classification drops 8.1 percentage points, F_multi classification drops 12.8pp, and odd/even drops 2.5pp. Simple tasks (addition, spatial, comparison) are unaffected. The general UV^T perturbation introduces a symmetric component in A that shifts eigenvalues from the imaginary axis (rotation) toward the real axis (scaling/contraction), creating gradient instability through the coupling between cross-entropy loss and $\exp(A)$.

6.2 Symmetric Regularization Restores Performance

The pade_symreg variant adds a penalty $\lambda_{\text{sym}} \|A_{\text{sym}}\|_F^2$ where $A_{\text{sym}} = (A + A^T)/2$. This directly suppresses the symmetric eigenvalue component without constraining the skew-symmetric (rotational) structure:

- Classification: $0.816 \to 0.889$ (full recovery to within error bars of skew_only)
- F_multi classification: $0.752 \to 0.880$ (full recovery)
- F_multi transfer: $0.750 \to 0.822$ (exceeds skew_only's 0.811)

The symmetric penalty confirms the gradient instability hypothesis: the degradation in full_pade comes specifically from the symmetric component creating real eigenvalues, not from the perturbation direction itself. When the symmetric component is regularized to small magnitude, the additional degree of freedom (symmetric scaling alongside rotation) is mildly beneficial.

6.3 Padé Approximation Is Sufficient

The matexp_symreg variant (exact `torch.linalg.matrix_exp`) closely matches pade_symreg across all conditions. The Cayley transform (Padé [1,1] approximation to the matrix exponential) is adequate for this architecture and task set. No measurable benefit accrues from the exact computation, which costs significantly more per forward/backward pass.

6.4 Geometric Interpretation

The eigenvalue decomposition of $A = S + K$ (symmetric + skew-symmetric) reveals:

- $\exp(K)$: imaginary eigenvalues $\to$ rotation, sinusoidal structure, volume-preserving
- $\exp(S)$: real eigenvalues $\to$ magnitude scaling, contraction/expansion

The symmetric regularization penalizes S directly ($\|A_{\text{sym}}\|_F^2 = \|S\|_F^2$). With sym_reg active: the rotational axis ($\exp(K)$) is fully free to learn, while the scaling axis ($\exp(S)$) is pushed toward zero but not eliminated. This occupies a specific position in the constraint space—distinct from the L1 penalty on UV^T (which targets the magnitude of the perturbation regardless of its symmetric/skew decomposition) and from structural constraints like the convolution recovery condition (which targets the *direction* of the off-diagonal blocks).

7 Discussion

7.1 Architecture IS the Geometric Prior: Definitive Evidence

The L1 results provide the strongest experimental evidence for the architecture-as-geometric-prior principle. The same architecture, with the same initial capacity at every layer, is trained on five tasks with identical hyperparameters. The only variable is the task itself. The result: a clean, interpretable hierarchy of geometric complexity that matches the tasks' structural properties:

- Tasks solvable by linear combination of features (addition, spatial) need the architectural prior alone
- Tasks requiring relational comparison (comparison) need minimal learned correction
- Tasks requiring global feature integration (odd/even) or fine-grained discrimination (classification) need progressively more learned departure from the prior

This hierarchy is discovered, not designed. The framework predicts that such a hierarchy *should* exist (because different tasks have different geometric complexity), but the specific mapping of tasks to complexity levels is an empirical contribution.

7.2 Automatic Architecture Discovery

The L1 findings validate a concrete approach to automatic architecture generation: start with a maximally expressive architecture (perturbation at every layer), apply L1 pressure, and read off the minimal architecture from the surviving structure. For a production system:

1. Train with L1 on perturbation parameters
2. Identify layers where perturbation is zeroed
3. Replace those layers with fixed canonical rotation (zero learned parameters)
4. Retrain the pruned architecture (optional—performance should match or improve)

For addition, this produces a pure routing + canonical rotation stack with *no learned mixer parameters*—the simplest possible architecture for the task, discovered automatically from the data's geometry.

7.3 Implications for Continual Learning

The flag manifold interpretation offers a geometric protocol for continual learning:

1. **Maintain the flag**: When adding a new task, keep all existing tasks in the loss so the optimizer preserves V_k while extending to $V_{k+1} \supset V_k$
2. **Extend, don't overwrite**: The optimizer should add new dimensions to the flag rather than rotating existing ones. The cumulative curriculum achieves this through the joint loss; structural approaches (e.g., growing the network or freezing existing parameters) could achieve it architecturally.
3. **Monitor meaning fractions**: μ_ℓ for existing tasks should not decrease when new tasks are added. A decrease signals that the new task's geodesic is being served at the expense of existing alignment.

The sequential replacement result demonstrates the cost of violating this protocol: catastrophic forgetting is not a mysterious failure mode but the predictable geometric consequence of overwriting a flag manifold's lower levels.

7.4 The Meaning Fraction as a Deployment Diagnostic

Spatial's $\mu = 0.138$ in the multi-task backbone makes a prediction: any task whose geodesic is nearly orthogonal to the backbone representation will generalize poorly under input perturbation, even when on-distribution performance appears excellent. This is a directly actionable diagnostic for deployed multi-task systems:

1. Compute μ_ℓ for each task at each layer
2. Tasks with low μ are vulnerable to distribution shift
3. Either: (a) add auxiliary tasks whose geodesics pull the backbone toward the low-μ task, or (b) add a dedicated subnetwork for the misaligned task

This diagnostic applies beyond the controlled MNIST setting to any multi-task system where tasks of different geometric character share a backbone. A low meaning fraction is a warning: the task may appear well-served on-distribution but will be fragile under perturbation or distribution shift.

7.5 Limitations

Scale. All experiments use 28×28 images, 5 tasks, state dimension 12, and ∼3,350 backbone parameters. The phenomena may not survive at transformer scale ($d = 4096$, billions of parameters). However, the quantities we measure (meaning fraction, L1-discovered complexity, flag convergence) are dimensionless ratios and structural patterns that have no inherent scale dependence.

Fixed architecture. All conditions use 5 layers of H_rotational. The theory predicts that optimal depth should vary per task (addition needs fewer layers than classification). We do not test variable-depth architectures here.

Early exit confound. The odd/even specialist early-exited at epoch 4 (accuracy $\geq$ 0.90 threshold). Its single-task performance (0.908) is a lower bound, not a converged value. Multi-task conditions ran for 50 full epochs. Comparisons involving odd/even should be interpreted accordingly.

Curriculum order. We test one curriculum order (spatial → cls → add → oe → cmp). The convergence result (same endpoint as F_multi) suggests order-invariance, but we have not verified this for all $5! = 120$ permutations.

8 Conclusion

We have presented experimental validation of the geometric prior framework's predictions about multi-task learning. Three contributions stand out:

1. L1 regularization discovers task complexity. Applied to the perturbation of a canonical rotation, L1 pressure automatically identifies which tasks are served by the architectural prior alone (addition, spatial: zero perturbation) and which require progressively more learned departure (comparison: one layer; odd/even: four layers; classification: all five). This is automatic architecture discovery: the data's geometry determines the minimal architecture, not a human designer.

2. Cumulative curriculum constructs a flag manifold. Progressive task addition with maintenance of all prior objectives builds a nested sequence of representational subspaces ($V_1 \subset V_2 \subset \cdots \subset V_5$) where each extension reinforces existing tasks. Keeping all prior tasks in the loss preserves the nesting and guarantees zero forgetting; the positive transfer (classification improving from 0.670 to 0.891 across phases) is the geometric signature of flag extension populating dimensions parallel to existing geodesics.

3. Coherence determines transfer, not rank. A backbone with high reservoir rank but incoherent content (from sequential overwriting) transfers worse than one with lower rank but coherent, mutually reinforcing meaning directions. The perpendicular reservoir enables transfer only when it contains directions parallel to the new task's geodesic.

These results establish measurable geometric quantities—meaning fraction, flag nesting, reservoir coherence—as diagnostics for multi-task representation quality, with direct implications for continual learning, automatic architecture generation, and any multi-task system where structurally diverse tasks share a learned representation.

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